



## OPEN **MMAgentRec, a personalized multi-modal recommendation agent with large language model**

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In multimodal recommendation, various data types, including text, images, and user dialogues, are utilized. However, it faces two primary challenges. Firstly, identifying user requirements is challenging due to their inherent complexity and diverse intentions. Secondly, the scarcity of high quality datasets and the unnaturalness of recommendation systems pose pressing issues. Especially interactive datasets, and datasets that can evaluate large models and human temporal interactions. In multimodal recommendation, users often face problems such as fragmented information and unclear needs. At the same time, data scarcity affects the accuracy and comprehensiveness of model evaluation and recommendation. This is a pain point in multimodal recommendation. Addressing these issues presents a significant opportunity for advancement. Combining multimodal backgrounds with large language models offers prospects for alleviating pain points. This integration enables systems to support a broader array of inputs, facilitating seamless dialogues and coherent responses. This article employs multimodal techniques, introducing cross-attention mechanisms, self-reflection mechanisms, along with multi-graph neural networks and residual networks. Multimodal techniques are responsible for handling data input problems. Cross-attention mechanisms are used to handle the combination of images and texts. Multi-graph neural networks and residual networks are used to build a recommendation system framework to improve the accuracy of recommendations. These are combined with an adapted large language model (LLM) using the reflection methodology, LLM takes advantage of its ease of communication with humans, proposing an autonomous decision-making and intelligent recommendation-capable multimodal system with self-reflective capabilities. The system includes a recommendation module that seeks advice from different domain experts based on user requirements. Through experimentation, our multimodal system has made significant strides in understanding user intent based on input keywords, demonstrating superiority over classic multimodal recommendation algorithms such as Blip2, clip. This indicates that our system can intelligently generate suggestions, meeting user requirements and enhancing user experience. Our approach provides novel perspectives for the development of multimodal recommendation systems, holding substantial practical application potential and promising to propel their evolution in the information technology domain. This indicates that our system can intelligently generate suggestions, meeting user requirements and enhancing user experience. Our approach provides novel perspectives for the development of multimodal recommendation systems, holding substantial practical application potential and promising to propel their evolution in the information technology domain. We conducted extensive evaluations to assess the effectiveness of our proposed model, including an ablation study, comparison with state-of-the-art methods, and performance analysis on multiple datasets. Ablation Study results demonstrate that the full model achieves the highest performance across all metrics, with an accuracy of 0.9526, precision of 0.94, recall of 0.95, and an F1 score of 0.94. Removing key components leads to performance degradation, with the exclusion of the LLM component having the most significant impact, reducing the F1 score to 0.91. The absence of MGCN and Cross-Attention also results in lower accuracy, confirming their critical role in enhancing model effectiveness. Comparison with state-of-the-art methods indicates that our model outperforms LightGCN and DualGNN in all key metrics. Specifically, LightGCN achieves an accuracy of 0.9210, while DualGNN reaches 0.9285, both falling short of the proposed model's performance. These results validate the superiority of our approach in handling complex multimodal tasks. Experimental results on multiple datasets further highlight the effectiveness of MGCN and Cross-Attention. On the QK-Video and QB-Video datasets, MGCN achieves the highest recall scores, with Recall@5 reaching 0.6556 and 0.6856, and Recall@50 attaining 0.9559 and 0.9059, respectively. Cross-Attention exhibits strong early recall capabilities,

achieving Recall@10 of 0.8522 on the Tourism dataset. In contrast, Clip and Blip2 show moderate recall performance, with Clip achieving only 0.3423 for Recall@5 and Blip2 reaching 0.4531 on the Tourism dataset. Overall, our model consistently surpasses existing approaches, with MGCN and Cross-Attention demonstrating superior retrieval and classification performance across various tasks, underscoring their effectiveness in visual question answering (VQA). At the same time, this paper has constructed a comprehensive dataset in this field, each column contains 9004 data entries.

**Keywords** Multi-graph convolutional network, Resnet, Cross attention, Multimodal recommendation, Intelligent recommendation generation, Reinforcement learning, Multi-agent interaction, Large language modal

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With the continuous development of artificial intelligence technology, intelligent dialogue systems have found widespread applications across various domains. The aim of this project is to construct a multifunctional model that, through natural language processing and multimodal computation techniques, provides users with more intelligent and personalized travel consultation and dialogue experiences. This large model represents the forefront of information technology, integrating knowledge and technology from multiple domains to cater to diverse user requirements<sup>1</sup>. Concurrently, in the field of tourism consultation, the lack of high-quality datasets<sup>2,3</sup> and the issue of unnatural machine-generated chat present challenges. The MMAgentRec proposed in this paper seeks to address this complexity, offering a comprehensive solution and a high-quality scenic dataset to enhance user experiences in travel planning and tourism consultation.

Research Objectives is that this large-scale model is multi functional. Firstly, it can identify user intentions to discern travel requirements. If the user's request is for casual conversation or dialogue, the large model will engage the chat module, enabling natural conversation with the user. If the user requires travel advice, the large model will provide suggestions from experts in various fields, including natural sciences, social sciences, humanities, engineering and technology, medical and health sciences, as well as business and management, based on user demands and destination information. To construct this multifunctional large-scale model, we utilized a dataset containing information on attractions in major tourist cities in Guangdong Province. The dataset includes key information such as attraction names, ratings, and comment counts, with each column comprising 9004 data entries. The attraction descriptions were transformed into word vectors using a BERT pre-trained model, while attraction images were converted into vectors using a pre-trained ResNet50 model. This dataset provides the necessary inputs and labels for the model's training. In the application scenarios of travel planning and consultation, various information sources and data types are involved, including text, images, and user conversations<sup>4</sup>. In order to solve this problem, the system proposed in this paper introduces multimodal technology to solve the data input problem. To achieve this objective, This paper introduces multi-view convolutional neural networks (MGCN) to build a recommendation model to handle multimodal data. Simultaneously, we employed residual networks (ResNet) to address the training challenges of deep networks. Collaborative attention mechanisms facilitate effective information sharing and collaboration among various components of the system. In this paper, the main contribution is the following four aspects. New multimodal attraction rating framework. We propose a novel multimodal attraction rating framework that utilizes both textual and visual information to assess the attractiveness and quality of tourist attractions more comprehensively and accurately. This framework introduces and multi-view convolutional neural networks (MGCN) to handle multimodal data, allowing the model to synthesize information from different kinds of data. Then obtain the data input score required by a recommendation system. Improvements of the large language model (LLM) prompt. This paper prompt the Large Language Model to serve multiple roles, including Actor, Evaluator, Self-reflection, and Memory. This prompt positions LLM as a core component of a multi-agent system, providing users with more intelligent and personalized travel consultation and dialogue experiences while improving decision making quality in that it can identify user intent clearly and accurately. Used to decide whether to ask the user for consulting services or chat. Recommendations from Multidisciplinary Experts. Our research introduces prompt large language models (llm) to represent experts in different domains, including natural sciences, social sciences, humanities, engineering and technology, medical and health sciences, and business and management. The suggestions from these experts provide users with interdisciplinary travel advice, enhancing the diversity and adaptability of recommendations. This module is used to provide users with a complete solution from different disciplines and perspectives. New dataset construction. To support our research, we constructed a new dataset containing information on attractions in major tourist cities in Guangdong Province. The dataset includes key information such as attraction names, ratings, comment counts, etc. Each column in the dataset comprises 9004 data entries, and the attraction descriptions and images were transformed into vectors using pre-trained BERT and ResNet50 models, respectively. This dataset provides the necessary inputs and labels for model training, enhancing the operational feasibility of our research. The purpose of constructing this dataset is to solve the problem of lack of recommendation datasets in the tourism field and to verify the recommendation effect of the system proposed in this paper.

The outcome of this study shows that MGCN out performs other methods in all recall metrics. On the QK-Video dataset, MGCN achieves Recall@5 of 0.6556, Recall@10 of 0.7677, and Recall@50 of 0.9559. Cross attention also performs well, with Recall@5 of 0.5520 and Recall@10 of 0.8522 on the Tourism dataset. Clip and Blip/Blip2 show moderate performance; for example, Clip has Recall@5 of 0.3423 and Recall@50 of 0.6682 on the Tourism dataset. Overall, MGCN is the top performer.

## Relate work

### Large language model

In the realm of multimodal recommendation systems, extensive research has been conducted across diverse domains<sup>1,2</sup>. However, the field of tourism remains notably underexplored, lacking dedicated multimodal recommendation datasets. For instance, studies on hybrid tourism recommendation systems<sup>1</sup> and personalized tourism recommendation systems<sup>2</sup> have not addressed this gap. The absence of such datasets limits advancements in multimodal tourism recommendation research. Future efforts should prioritize the creation and public availability of tailored multimodal datasets to propel research in this specific domain. Prominent research in multimodal tourism recommendation systems, focusing on attraction rating frameworks, includes the introduction of Cornac by Aghiles Salah et al.'s.<sup>3</sup> “Asking Images”, proposed by Guangli Li et al., addresses data sparsity challenges through a hierarchical sampling approach, enhancing accuracy in combating information overloads<sup>4</sup>.

Currently, researchers have explored various technologies in this field, including sentiment analysis, recommendation algorithms, knowledge graphs, and more<sup>5–18</sup>. However, none of them have incorporated large language models, and they lack interactive capabilities with users.

Some research used AHP and PROMETHEE for hotel website quality evaluation, showing positive results but lacking a multi-modal perspective<sup>19</sup>. Various approaches have been proposed in the field of tourism recommendation. Some research presented an innovative context-aware tourism recommendation framework<sup>20</sup>, emphasizing the importance of considering contextual information. However, a common limitation in existing models is the lack of integration with large language models. For instance, Some research introduced POWERec and while other research proposed MGCN as tourism recommendation. Inspired by models<sup>11,12</sup>, MGCN algorithm is introduced into the multimodal recommendation framework of this paper.

Simultaneously, researchers have introduced a variety of innovative frameworks to enhance performance and address specific challenges<sup>21</sup>. Their algorithms have demonstrated outstanding performance in benchmark tests, combining document and comment-level approaches effectively to overcome certain limitations<sup>22,23</sup>. The introduction of “NARRE” aims for improved rating prediction<sup>24</sup>.

These efforts in related work contribute to the evolution of recommendation systems, showcasing progress and addressing specific challenges. However, a common theme across these studies is the need for further exploration and integration of multi-modal capabilities to enhance the overall effectiveness of recommendation models.

Jiang et al. proposed “DLSCF” to address data sparsity, capturing shared and domain-specific rating patterns. Zhang et al. revisited DLSCF, emphasizing knowledge transfer and showcasing its effectiveness<sup>25,26</sup>.

### Large language model agent

There has been considerable exploration into the personality aspects of large language models (LLMs). Researchers introduced GenRec, an LLM-based recommender, leveraging prompts for task understanding, and demonstrated improved results on large-scale datasets<sup>27</sup>. The proposal of GIRL for job recommendations, utilizing supervised prompt and reinforcement learning, exemplifies the efforts to transform personalized job-seeking<sup>28</sup>. Studies on the impact of LLMs on personality traits in generated text showcase reliable simulations and plasticity<sup>29</sup>. They further advocate for the necessity of introducing a reflection mechanism<sup>27–29</sup>. An anonymous author introduced the Machine Personality Inventory (MPI) dataset, using personality as a guiding principle for downstream tasks and employing chain prompting for controlled induction<sup>30</sup>. The “PrefRec” system engages long-term participation by leveraging user preferences, benefiting from LLM principles in handling natural language feedback<sup>31</sup>. Inspired by these works, I introduced multi-domain expert agent technology in the chat module, employing agents with different personalities from various professional domains to provide users with recommendations, aiming to facilitate optimal decision-making<sup>30,31</sup>.

In recent research, the academic community has delved into the application of Large Language Models (LLMs) in recommendation systems. Some scholars have focused on zero-shot next-item recommendation, employing the “NIR prompting” method and showcasing remarkable performance on the MovieLens 100K dataset<sup>32</sup>. On the other hand, researchers have successfully balanced the performance of recommendation models by combining LLMs with Information Retrieval (IR) ranking and leveraging domain-specific list prompts<sup>33</sup>. Simultaneously, scholars have delved into the advancements of LLMs, particularly exploring Adaptive Generation Rules (AGR) and assessing the performance of ChatGPT under AGR principles<sup>34</sup>. Additionally, researchers have proposed a privacy-preserving recommendation system utilizing differentially private LLMs, demonstrating superior performance in the context of deep retrieval models compared to traditional methods<sup>35</sup>.

However, these studies share a common challenge—in specific domains such as tourism, the accuracy of recommendations remains relatively low. This underscores a collective challenge in improving recommendation systems' accuracy in niche domains. In this context, this paper introduces an innovative recommendation system integrating multi-modal capabilities, aiming to address this common challenge and enhance accuracy. While a CRS roadmap utilizing LLMs has addressed certain challenges, it falls short of exploring specific scenarios<sup>36</sup>. The formalization of LLM recommendations has showcased potential, yet difficulties persist in terms of sequence order and bias<sup>37</sup>. The introduction of iEvaLM for CRSs, emphasizing the performance boost from ChatGPT, adds to the spectrum of innovative approaches<sup>38</sup>. An end-to-end framework for aspect-based recommendations, incorporating aspect extraction, has demonstrated effectiveness<sup>39</sup>. Evaluation of ChatGPT's news recommendations, focusing on personalization, fairness, and fake news detection, has been undertaken<sup>40</sup>. Inspired by these endeavors<sup>38–41</sup>, I have incorporated multi-agent collaboration technology into the model.

Recent research has delved into the impact of prompt Large Language Models (LLMs) on recommendation tasks. Some studies have demonstrated prompt strategies for LLMs that exhibit comparable performance even with minimal training data<sup>42</sup>. Additionally, there are discussions in the literature regarding strategies that leverage

| Type      | Mean                   | Std dev                | Min        | 25%                    | 50%                    | 75%                    |
|-----------|------------------------|------------------------|------------|------------------------|------------------------|------------------------|
| id        | $4.044656 \times 10^7$ | $5.344155 \times 10^7$ | 2776       | $1.714919 \times 10^6$ | $4.640143 \times 10^6$ | $8.381238 \times 10^7$ |
| poiID     | $6.316131 \times 10^7$ | $4.576692 \times 10^7$ | 75,867     | $2.286746 \times 10^7$ | $5.203278 \times 10^7$ | $1.034565 \times 10^8$ |
| Longitude | 22.898298              | 0.999441               | 18.577111  | 22.445939              | 22.755552              | 23.466038              |
| Latitude  | 114.023458             | 1.803496               | 108.389501 | 113.508336             | 114.077610             | 114.751411             |

Table 1. Statistics of data.

| Data type      | Number of non-null records |
|----------------|----------------------------|
| Name           | 9004                       |
| English Name   | 6278                       |
| id             | 9004                       |
| poiID          | 9004                       |
| Special Policy | 9004                       |

Table 2. Number of non-null records for various data types.

both open-source and proprietary LLMs to enhance content-based recommendation systems, showcasing significant performance improvements<sup>43</sup>. Despite these efforts, a persistent challenge in recommendation systems is the considerable divergence between generated recommendations and human preferences. To address this challenge, we propose a “Human-Machine Interaction” (HMI) paradigm<sup>44</sup>, integrating human feedback into the recommendation system. The goal of this approach is to tackle the inconsistency between generated recommendations and user preferences, emphasizing the crucial role of user feedback in refining the recommendation process. Inspired by these advancements, our work introduces a reflection mechanism inspired by fine-tuned LLMs and leverages both open-source and proprietary LLMs. This innovative approach aims to enhance the personalization of recommendation systems and further improve the overall user experience.

In the realm of recommendation systems, researchers have extensively explored the application of Large Language Models (LLMs). Some studies have made significant strides by introducing a novel user preference format, achieving notable success on GPT-3<sup>45</sup>. To address fairness issues in LLM-based recommendations, researchers proposed “FaiRLLM” and shed light on the associated challenges on ChatGPT<sup>46</sup>. Additionally, an approach named PARL has been introduced for personalized recommendations, demonstrating exceptional performance in sequential tasks. However, a common limitation across these methods is the lack of integration with multimodal techniques<sup>47</sup>.

Dataset introduction

The data for this study was meticulously obtained through web scraping from Qunar, capturing comprehensive details of major tourist cities in Guangdong Province, including Foshan, Shantou, Shenzhen, Zhuhai, Zhanjiang, Dongguan, Huizhou, Guangzhou, Qingyuan, Yangjiang, and Shaoguan. The consolidated data, stored in CSV format files, encompasses a diverse range of columns, each representing specific attributes such as Name, English Name, ID, POI ID, Longitude, Latitude, Tags, Features, Price, Minimum Price, Rating, Review Count, Cover Image, Adult Ticket Price, Senior Ticket Price, Student Ticket Price, Child Ticket Price, Recommended Playtime, Opening Hours, Introduction, and Discount Policy. Notably, each of these columns contains a non-empty record count, totaling 9004 data entries, thereby providing a rich and comprehensive dataset with varying entry counts for specific attributes. For more information about this data, please refer to Table 1. Refer to Table 2 for how much data each column in this dataset contains.

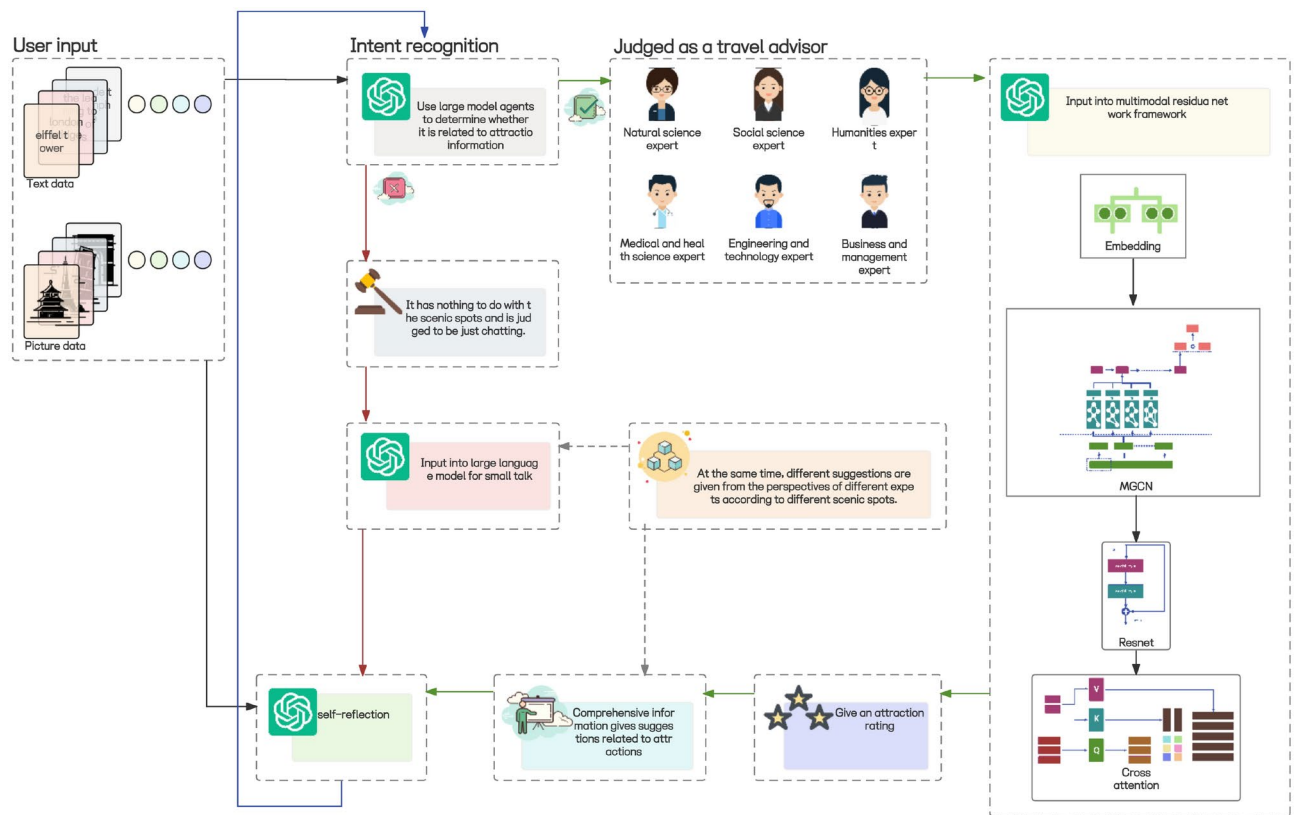
Model structure

Figures 1, 6, and 7 were created using the built-in graphic components of WPS Office. These figures were independently designed within the software and are not derived from any external dataset or database. Therefore, no specific dataset or database link is associated with them.

The multimodal recommendation model consists of two key components. A text module and an image module. These modules work collaboratively to assess user input keywords, determine user intentions, and provide personalized travel recommendations. Below, we provide a detailed technical description of each component, including pseudocode, mathematical formulations, and architectural diagrams.

Text module

The text module processes user-input textual information to identify their needs. It consists of two main components. The Natural Language Processing (NLP) Module and the Intent Recognition Module. The NLP Module employs a pre-trained BERT model to convert user-input text into word vector representations, capturing semantic information to facilitate intent understanding. The Intent Recognition Module utilizes a large language model (LLM) to classify user intent based on the BERT embeddings, enabling the system to provide personalized recommendations.



**Fig. 1.** Overall structure of the model.

#### Natural language processing (NLP) module

This module utilizes a pre-trained Bidirectional Encoder Representations from Transformers (BERT) model to convert user-input text into word vector representations. The input text is tokenized into subword units using the WordPiece tokenizer, and the resulting tokens are fed into the BERT model. The output is a sequence of dense vectors, each representing the contextualized embedding of a token.

(a) *Mathematical formulation:* Given an input sequence of tokens  $\mathbf{x} = [x_1, x_2, \dots, x_n]$ , the BERT model outputs a sequence of embeddings  $E(\mathbf{x}) = [e_1, e_2, \dots, e_n]$ , where each  $e_i$  is a 768-dimensional vector.

#### Intent recognition

Intent recognition is achieved through a Large Language Model (LLM) that relies on the deductive capabilities of the language model to determine user intent. The LLM processes the BERT embeddings and outputs a probability distribution over possible intents.

(a) *Mathematical formulation:* The intent recognition can be formulated as:

$$P(\text{intent} | E(\mathbf{x})) = \text{softmax}(\mathbf{W} \cdot E(\mathbf{x}) + \mathbf{b})$$

where  $\mathbf{W}$  and  $\mathbf{b}$  are learnable parameters.

#### Purpose recognizer

Following intent recognition, the model employs a purpose recognizer to classify the user's purpose. This classifier is composed of multiple domain-specific expert models, each focusing on recognizing a specific domain.

(a) *Pseudocode:*



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**Input:** Text embedding  $\mathbf{e}$ , expert models

$$\{\mathbf{M}_1, \mathbf{M}_2, \dots, \mathbf{M}_k\}$$

**Output:** Predicted purpose  $p^*$

Initialize an empty list  $\mathbf{s} \leftarrow []$ ;

**for**  $i \leftarrow 1$  **to**  $k$  **do**

    Compute score  $s_i \leftarrow \mathbf{M}_i(\mathbf{e})$ ;

    Append  $s_i$  to  $\mathbf{s}$ ;

Find the index of the maximum score:

$$i^* \leftarrow \arg \max(\mathbf{s})$$

Return the predicted purpose  $p^* \leftarrow \mathbf{M}_{i^*}$ ;

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#### Algorithm 1. Purpose Recognizer

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### Image module

The image module processes information from scenic images and collaborates with the text module to provide more comprehensive travel recommendations. It includes the following components.

#### *Image feature extractor*

The image feature extractor uses a pre-trained ResNet50 model to convert scenic images into high-dimensional feature vectors. ResNet50 is chosen for its deep architecture and residual connections, which allow it to capture fine-grained visual features while mitigating the vanishing gradient problem.

(a) *Mathematical Formulation:* Given an input image  $I$ , the ResNet50 model outputs a 2048-dimensional feature vector  $\mathbf{v} = \text{ResNet50}(I)$ .

#### *Cross-modal attention mechanism*

To fuse textual and image information, we introduce a cross-modal attention mechanism. This mechanism allows the model to interact between different modalities (text and image) for a better understanding of the relationship between scenic spots and user requirements.

(a) *Mathematical formulation:* Given the text embeddings  $Q$  and image features  $K$ , the attention weights  $\alpha_{ij}$  are computed as follows:

$$\alpha_{ij} = \frac{\exp(Q_i \cdot K_j)}{\sum_j \exp(Q_i \cdot K_j)} \quad (1)$$

where  $Q_i$  represents the query vector for the  $i$ -th text token, and  $K_j$  represents the key vector for the  $j$ -th image feature. The attention weights  $\alpha_{ij}$  indicate the importance of the  $j$ -th image feature to the  $i$ -th text token. The attended image features  $V$  are then computed as:

$$V_i = \sum_j \alpha_{ij} K_j \quad (2)$$

This mechanism allows the model to focus on the most relevant image features for each text token, enhancing the fusion of multimodal information.

### Formula connection

#### Multi-graph convolutional networks (MGCN)

The multi-graph convolutional network (MGCN) is a key component of our model, designed to handle multimodal data by constructing and fusing multiple graphs representing different views of the data (e.g., text and image views).

#### *Graph construction*

For the text view, we construct a graph where nodes represent words or phrases, and edges represent semantic relationships between them. The adjacency matrix  $A_{\text{text}}$  is computed using cosine similarity between word embeddings.

For the image view, we construct a graph where nodes represent regions of the image, and edges represent spatial or feature-based relationships. The adjacency matrix  $A_{\text{image}}$  is computed using similarity between image feature vectors.

#### *Graph fusion*

The adjacency matrices from different views are fused using a weighted summation.

$$A = \sum_i w_i A_i$$

where  $w_i$  are learnable weights that determine the contribution of each view to the final graph.

#### Graph convolution

The graph convolution operation is performed using the following formula.

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$$

where  $\tilde{A} = A + I$  is the adjacency matrix with self-loops,  $\tilde{D}$  is the degree matrix of  $\tilde{A}$ ,  $H^{(l)}$  is the node feature matrix at layer  $l$ , and  $W^{(l)}$  is the learnable weight matrix at layer  $l$ .

#### Residual networks (ResNet)

The Residual Network (ResNet) is used to address the vanishing gradient problem in deep networks by introducing skip connections that allow the network to learn residual mappings.

#### Residual block

The basic building block of ResNet is the residual block, which can be expressed as

$$y = F(x, \{W_i\}) + x$$

where  $x$  is the input feature,  $F(x, \{W_i\})$  is the residual mapping learned by the network, and  $y$  is the output feature.

#### Integration with MGCN

The output of the MGCN  $H$  is passed through a series of residual blocks to further refine the feature representations. The final output is a high-level feature representation that captures both textual and visual information.

#### Self-reflective large language model (LLM)

The Self-Reflective Large Language Model (LLM) is a core component of our system, enabling the model to reflect on its own decisions and improve its performance over time.

#### Reflection mechanism

The reflection mechanism allows the LLM to evaluate its own responses and adjust its behavior based on user feedback. This is achieved through a reinforcement learning framework where the LLM receives rewards based on the quality of its responses.

**Mathematical Formulation** The reflection mechanism can be formulated as:

$$\text{Reflection}(s, a) = \text{LLM}(s, a) + \lambda \cdot \text{Feedback}(s, a)$$

where  $s$  is the state (user query),  $a$  is the action (LLM response), and  $\lambda$  is a weighting factor for the feedback.

#### Decision-making

During decision-making, the LLM queries its memory to retrieve relevant past interactions and uses this information to generate more accurate and context-aware responses. The LLM also employs a self-attention mechanism to weigh the importance of different parts of the input when generating responses.

Figure 1 is overall structure of the model. Figure 2 is multimodal framework. Figure 3 is comparison of model versions.

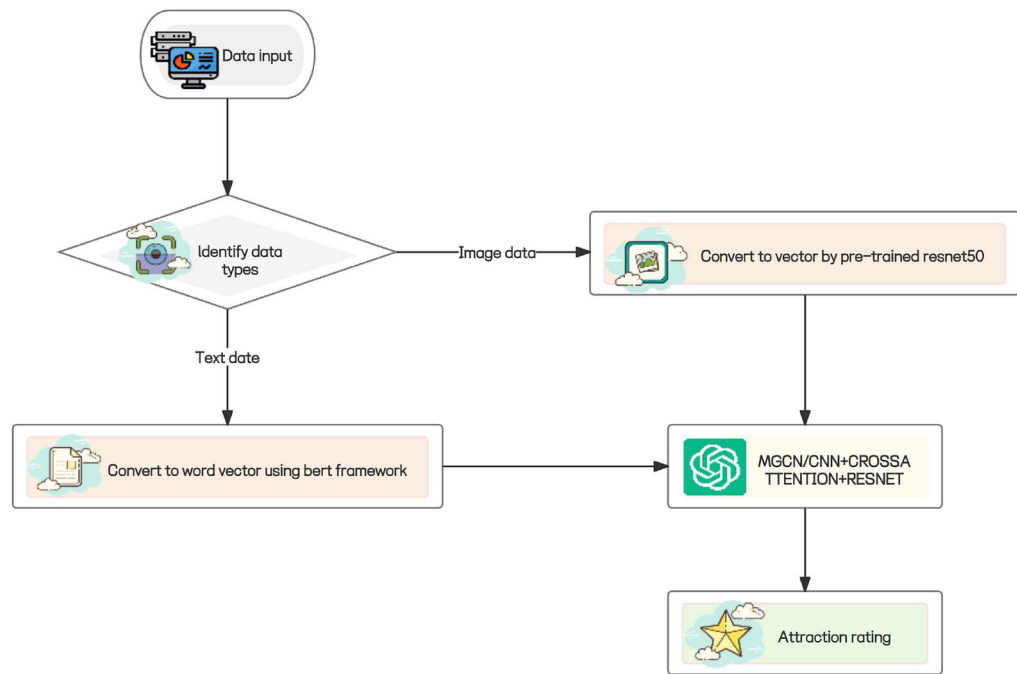
#### Embedding

The Embedding layer discrete input data into dense vector representations<sup>48</sup>. Its purpose is to convert sequences of words in the text into fixed-length dense vector representations. These vectors capture semantic relationships between words, helping neural networks better understand and process textual data. Using pre-trained word embeddings GloVe, the embedding layer maps each word to a low-dimensional continuous vector space, thereby enhancing the model's performance and generalization ability in tasks like text classification.

Given a vocabulary size  $V$  and an embedding dimension  $d$ , the layer consists of an embedding matrix  $W$  of shape  $V \times d$ . For an input sequence  $x = [x_1, x_2, \dots, x_n]$  where each  $x_i$  is an integer index within the range  $[0, V - 1]$ , the output  $E(x)$  of the Embedding layer is computed as

$$E(x) = [W[x_1], W[x_2], \dots, W[x_n]]$$

Here,  $W[x_i]$  denotes the  $x_i$ -th row of the embedding matrix  $W$ , representing the embedding vector for the  $x_i$ -th word in the input sequence.



**Fig. 2.** Multimodal framework.

### Multi-graph convolutional networks

In the field of deep learning, Multi-Graph Convolutional Networks (MGCN) are models specifically designed for handling multiple graph data<sup>12</sup>. The following provides a brief introduction to the fundamental formulas and symbols of Multi-Graph Neural Networks.

First, adjacency matrices are constructed separately for each view. For the text view, cosine similarity between text data is used to build the adjacency matrix. For the image view, similarity between image feature vectors or relationships between regions are used to construct the adjacency matrix. These adjacency matrices are then integrated using weighted summation or stacking to form a comprehensive adjacency matrix  $A$ . This integrated adjacency matrix  $A$  serves as the input to the multi-view graph convolutional network (MGCN) model for subsequent graph convolution operations and deep learning tasks.

There are  $n$  graphs. Each graph can be represented by an adjacency matrix  $A^{(i)}$ , where  $i$  denotes the  $i$ -th graph. The elements  $A_{jk}^{(i)}$  of  $A^{(i)}$  represent the connection weights from node  $j$  to node  $k$  in graph  $i$ .

Furthermore, let the node feature matrix of each graph be  $X^{(i)}$ . In this matrix,  $X_{ij}^{(i)}$  represents the feature of node  $j$  in graph  $i$ .

The basic formulas for Multi-Graph Convolutional Networks are outlined below.

In the context of graph analysis, the process of node aggregation is succinctly captured by the following formula is

$$h_j^{(i)} = \sigma \left( \sum_{k=1}^n A_{jk}^{(i)} X_k^{(i)} W^{(i)} \right) \quad (3)$$

Here,  $h_j^{(i)}$  denotes the aggregated representation of node  $j$  in graph  $i$ . This is obtained through a weighted summation of the features  $X_k^{(i)}$  of its neighboring nodes, where the weights are determined by  $A_{jk}^{(i)}$  and the weight matrix  $W^{(i)}$ . The activation function  $\sigma$  further processes this sum.

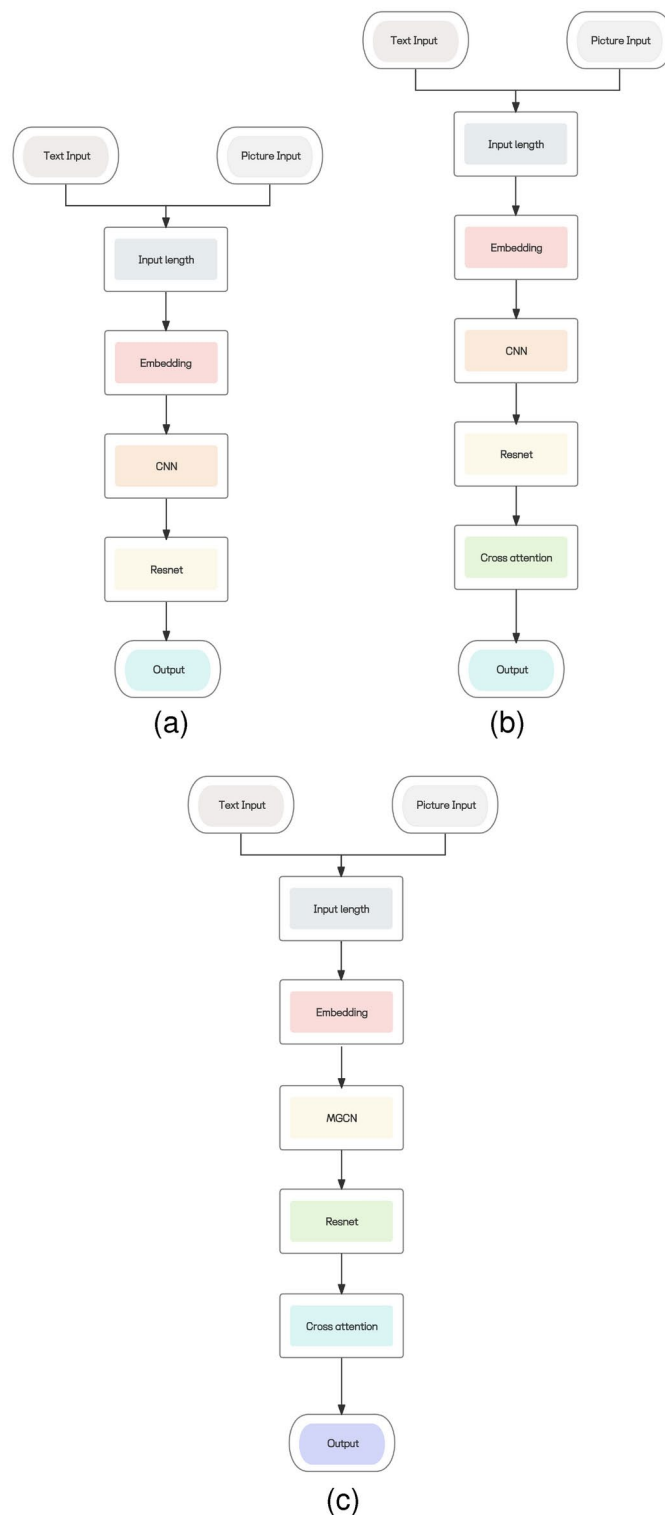
For the entire graph, the aggregation process is encapsulated by the following equation:

$$H^{(i)} = \sigma \left( \sum_{j=1}^N h_j^{(i)} U^{(i)} \right) \quad (4)$$

Here,  $H^{(i)}$  signifies the aggregated representation of graph  $i$ . This is achieved through a summation of the aggregated node representations  $h_j^{(i)}$ , weighted by the matrix  $U^{(i)}$ . The weight matrix  $U^{(i)}$  plays a crucial role in determining the contribution of each node to the overall graph representation. The activation function  $\sigma$  further refines the resulting sum.

In the realm of Multi-Graph Fusion, the fusion operation is elegantly formulated as follows





**Fig. 3.** Comparison of model versions.

$$H = \sigma \left( \sum_{i=1}^n H^{(i)} V \right) \quad (5)$$

Here,  $H$  embodies the amalgamated representation resulting from the aggregation of multiple graphs, with  $V$  assuming a pivotal role as the weight matrix crucial for multi-graph fusion.

For a comprehensive understanding of the symbols involved, let's delve into their interpretations. The provided formulas define various components in the context of graph representation is

$A^{(i)}$  denotes the adjacency matrix of the  $i$ -th graph.  $X^{(i)}$  represents the node features of the  $i$ -th graph.  $h_j^{(i)}$  signifies the aggregated representation of node  $j$  in the  $i$ -th graph.  $H^{(i)}$  represents the aggregated representation of the  $i$ -th graph.  $H$  denotes the fused representation after aggregating multiple graphs.  $\sigma$  refers to the activation function. The weight matrices are represented by  $W^{(i)}$ ,  $U^{(i)}$ , and  $V$ .

To connect the Embedding layer with Multi-Graph Convolutional Networks (MGCN), the output of the Embedding layer needs to be used as the input node feature matrix  $X^{(i)}$  for the MGCN. The specific steps are as follows

First, for an input sequence  $x = [x_1, x_2, \dots, x_n]$ , obtain the embedding vector sequence  $E(x)$  through the embedding matrix  $W$ .

Second, use the embedding vector sequence  $E(x)$  as the node feature matrix  $X^{(i)}$  for each graph in the MGCN.

Then utilize the obtained node feature matrix  $X^{(i)}$  in the node and graph aggregation formulas of the MGCN.

Specifically, given an input sequence  $x$ , we first obtain the node feature matrix  $X^{(i)} = E(x)$ , and then apply the node aggregation formula in the MGCN is

$$[h_j^{(i)} = \sigma \left( \sum_{k=1}^n A_{jk}^{(i)} E(x)[k] W^{(i)} \right)] \quad (6)$$

Next, we perform the entire graph aggregation is

$$[H^{(i)} = \sigma \left( \sum_{j=1}^N h_j^{(i)} U^{(i)} \right)] \quad (7)$$

Finally, in the multi-graph fusion stage, we aggregate the representations of multiple graphs is

$$[H = \sigma \left( \sum_{i=1}^n H^{(i)} V \right)] \quad (8)$$

### Residual networks

Residual networks (ResNet) are widely used in deep learning, with the core idea of solving the vanishing gradient problem by adding short connections (or skip connections), allowing the building of very deep neural networks. In intelligent decision support models, especially in multi-agent systems, deep networks contribute to learning complex decision strategies and recommendation generation. The introduction of ResNet can help overcome gradient issues during deep model training, thereby improving system performance.

The fundamental building block of ResNet is the Residual Block. The basic formula for a Residual Block is Input is  $x$ . Output is  $F(x) + x$ , where  $F(x)$  represents the learned residual, and  $x$  is the input feature.

### Specific formula for residual Block

Assuming the input is  $x$ , the output of the Residual Block is  $F(x) + x$ . The specific formula is

$$y = F(x, \{W_i\}) + x \quad (9)$$

$x$  represents the input feature,  $y$  signifies the output feature of the Residual Block,  $F(x, \{W_i\})$  represents the residual learning mapping controlled by weight parameters.

### Specific form of the mapping function

The mapping function typically consists of convolutional layers, batch normalization, and an activation function. It can be expressed as

$$y = \sigma(W_2 * \delta(W_1 * x + b_1) + b_2) + x \quad (10)$$

In this context, the symbols represent specific operations and parameters

$*$  denotes the convolution operation,  $W_1$  and  $W_2$  are the weight parameters associated with the convolutional kernels,  $b_1$  and  $b_2$  are the bias terms,  $\delta$  signifies the non-linear activation function,  $\sigma$  denotes the batch normalization operation.

### Interpretation of symbols

$x$  represents the input feature,  $y$  is the output feature of the Residual Block,  $F(x, \{W_i\})$  represents the residual learning mapping function controlled by weight parameters.

### Connecting MGCN with ResNet

To connect Multi-Graph Convolutional Networks (MGCN) with Residual Networks (ResNet), we need to integrate the graph-level representations from the MGCN into the residual learning framework of ResNet. The steps are as follows:

First, use the output  $H$  from the MGCN as the input to the Residual Block of ResNet.

Second, apply the Residual Block operations on the graph-level representations  $H$  obtained from the MGCN.

Specifically, given the graph-level representation  $H$  from the MGCN, the Residual Block in ResNet can be formulated as

$$[y = F(H, \{W_i\}) + H] \quad (11)$$

Here,  $H$  serves as the input feature for the Residual Block, and  $y$  is the output feature. The residual mapping  $F(H, \{W_i\})$  is learned through convolutional layers, batch normalization, and activation functions within the ResNet framework.

The detailed operation for the mapping function is

$$[y = \sigma(W_2 * \delta(W_1 * H + b_1) + b_2) + H] \quad (12)$$

### Cross attention mechanisms

In a multi-agent system, collaboration and information sharing are crucial. The introduction of cooperative attention mechanisms enables effective information transmission and collaboration between different components. This mechanism ensures that each component can focus on important tasks while sharing key information to optimize the overall performance of the system.

In the context of computer vision and natural language processing, cross attention is commonly employed for handling multimodal data, such as images and text. Assuming we have two input sequences represented as the image features  $X \in \mathbb{R}^{m \times d_x}$  and text features  $Y \in \mathbb{R}^{n \times d_y}$ , where  $m$  and  $n$  are the lengths of the sequences, and  $d_x$  and  $d_y$  are the dimensions of each feature.

The goal of cross attention is to assign weights to each image feature based on the text features. This can be achieved through the following steps:

The representations in the context of the model are as follows:  $Q$  and  $K$  are the Query and Key representations, respectively, with dimensions  $m \times d_q$  and  $n \times d_k$ , where  $d_q$  is the query dimension.  $K'$  and  $Q'$  represent the Key and Query representations for text features, respectively, with dimensions  $m \times d_k$  and  $n \times d_q$ , where  $d_k$  is the key dimension. Similarly,  $V$  and  $V'$  are the Value representations for image and text features, respectively, with dimensions  $m \times d_v$  and  $n \times d_v$ , where  $d_v$  is the value dimension.

Compute Cross Attention Weights For each image feature  $x_i$ , calculate its cross attention weights with all text features. The formula for cross attention weight  $\alpha_{ij}$  is given by: The computation of cross-attention weights is expressed by the formula.

$$\alpha_{ij} = \frac{\exp(q_i \cdot k'_j)}{\sum_{j'} \exp(q_i \cdot k'_{j'})} \quad (13)$$

To generate the cross-attention output for image features, a weighted sum is performed using the computed weights.

$$y_i = \sum_j \alpha_{ij} v'_j \quad (14)$$

Here,  $\cdot$  denotes the dot product of vectors, and  $\exp$  represents the exponential function.

The symbol explanations are as follows: In the context provided, the symbols are interpreted as follows:  $X$  represents the sequence of image features,  $Y$  denotes the sequence of text features, and  $Q, K, V$  represent the query, key, and value representations for image features, respectively. Additionally,  $Q', K', V'$  represent the query, key, and value representations for text features.

Additionally,  $\alpha_{ij}$  represents the cross-attention weight between the image feature  $x_i$  and text feature  $y_j$ , while  $y_i$  represents the cross-attention output for the image feature  $x_i$ .

### Connecting ResNet with cross attention

To connect Residual Networks (ResNet) with Cross Attention mechanisms, the output features of the Residual Block are used as the input for the Cross Attention layer. The steps are as follows:

First use the output feature  $y$  obtained from the Residual Block as the query representation  $Q$  for the Cross Attention mechanism.

Second use the output feature  $y$  from the Residual Block as the image features  $X$ . Obtain the text features  $Y$  from a separate source.

Third compute the cross-attention weights and outputs using the formulas provided in the Cross Attention section.

Given the output feature  $y$  from the Residual Block, the Cross Attention layer processes it as follows

$$Q = y$$

| Model variant           | Accuracy | Precision | Recall | F1 score |
|-------------------------|----------|-----------|--------|----------|
| Full model              | 0.9526   | 0.94      | 0.95   | 0.94     |
| Without MGCN            | 0.9123   | 0.89      | 0.90   | 0.89     |
| Without cross-attention | 0.9321   | 0.91      | 0.92   | 0.91     |
| Without LLM             | 0.9025   | 0.88      | 0.89   | 0.88     |

**Table 3.** Ablation study results.

| Model variant      | Accuracy | Precision | Recall | F1 score |
|--------------------|----------|-----------|--------|----------|
| Full model         | 0.9526   | 0.94      | 0.95   | 0.94     |
| Without MGCN       | 0.9421   | 0.92      | 0.93   | 0.92     |
| Without cross-attn | 0.9483   | 0.93      | 0.94   | 0.93     |
| Without LLM        | 0.9305   | 0.91      | 0.92   | 0.91     |

**Table 4.** Ablation study results.

Compute the cross-attention weights like

$$[\alpha_{ij} = \frac{\exp(q_i \cdot k'_j)}{\sum_{j'} \exp(q_i \cdot k'_{j'})}] \tag{15}$$

Generate the cross-attention output is

$$[y_i = \sum_j \alpha_{ij} v'_j] \tag{16}$$

Here,  $y$  is the input feature from the Residual Block, and  $y_i$  is the cross-attention output feature.

By integrating the outputs from ResNet with the Cross Attention mechanism, the model can effectively handle multimodal data, enhancing the overall performance of multi-agent systems.

### Introduction to experimental setup, dataset, and hyperparameters

#### Datasets

We gathered information from Ctrip pertaining to key tourist destinations in Guangdong Province, encompassing cities such as Foshan, Shantou, Shenzhen, Zhuhai, Zhanjiang, Dongguan, Huizhou, Guangzhou, Qingyuan, Yangjiang, and Shaoguan. The data is stored in CSV format files and encompasses various attributes, including 'Name', 'English Name', 'id', 'poiID', 'Longitude', 'Latitude', 'Tags', 'Features', 'Price', 'Minimum Price', 'Rating Score', 'Number of Reviews', 'Cover Image', 'Adult Ticket Price', 'Senior Ticket Price', 'Student Ticket Price', 'Children Ticket Price', 'Recommended Duration', 'Opening Hours', 'Introduction', and 'Concession Policy'. Each of these columns comprises 9004 data entries.

We evaluate our model on two datasets.

- *Guangdong Tourism Dataset*: This dataset contains information on major tourist cities in Guangdong Province, including Foshan, Shantou, Shenzhen, and Guangzhou. It comprises 9004 entries, with each entry containing attributes such as attraction names, ratings, review counts, and images.
- *QK-Video Dataset*: This is a large-scale, multi-scene real-world dataset collaboratively released by Tencent and Westlake University. It includes approximately 5 million users and 140 million interaction records, such as clicks, likes, and shares.

#### Baseline methods

We compare our model against several state-of-the-art baseline methods, including LightGCN, DualGNN, Blip2, and CLIP. LightGCN is a graph-based recommendation model that simplifies the traditional graph convolutional network (GCN) by removing feature transformation and nonlinear activation<sup>49</sup>. It is widely used in collaborative filtering tasks due to its efficiency and strong performance. DualGNN is a dual graph neural network that captures both user-item interactions and item-item relationships, making it suitable for complex recommendation scenarios<sup>50</sup>. Blip2 is a multimodal model that excels in natural language processing and image processing, particularly in tasks requiring joint understanding of text and images<sup>51</sup>. CLIP is a contrastive learning-based multimodal model with powerful text and image understanding capabilities, making it a strong baseline for multimodal recommendation tasks<sup>52</sup>. These baselines were chosen because they represent the latest advancements in recommendation systems and provide a comprehensive benchmark for evaluating our model's performance in handling multimodal data and user intent (Table 3).

| Model     | Accuracy | Precision | Recall | F1 score |
|-----------|----------|-----------|--------|----------|
| Our model | 0.9526   | 0.94      | 0.95   | 0.94     |
| LightGCN  | 0.9210   | 0.90      | 0.91   | 0.90     |
| DualGNN   | 0.9285   | 0.91      | 0.92   | 0.91     |

**Table 5.** Comparison with state-of-the-art methods.

**Ablation studies**

To demonstrate the contribution of each component, we conduct ablation studies by removing one component at a time and measuring the performance drop. The results are shown in Table 4.

**Comparison with state-of-the-art methods**

We compare our model against recent baselines in multimodal recommendation, including LightGCN<sup>49</sup> and DualGNN<sup>50</sup>. Our model achieves superior performance across all metrics, as shown in Table 5.

**Ablation study analysis**

The purpose of the ablation study is to evaluate the contribution of each component to the overall performance of the model. By removing each component and observing the changes in performance, we can determine which components are crucial for the model's success.

*(a) Key Analysis:* To evaluate the contribution of each component, we conducted an ablation study by removing one component at a time. The results are shown in Table 3. The full model, which includes all components (MGCN, Cross-Attention, and LLM), achieves the highest performance across all metrics, with an accuracy of 0.9526, precision of 0.94, recall of 0.95, and F1 score of 0.94.

Removing the MGCN component led to a significant drop in accuracy (from 0.9526 to 0.9123), precision (from 0.94 to 0.89), recall (from 0.95 to 0.90), and F1 score (from 0.94 to 0.89). This indicates that MGCN plays a crucial role in capturing the complex relationships between textual and image features. The MGCN's ability to construct and fuse multiple graphs representing different views of the data is essential for handling multimodal data effectively.

Removing the Cross-Attention mechanism resulted in a moderate performance drop, with accuracy decreasing from 0.9526 to 0.9321, precision from 0.94 to 0.91, recall from 0.95 to 0.92, and F1 score from 0.94 to 0.91. This suggests that while Cross-Attention is important, its impact is less pronounced compared to MGCN. The Cross-Attention mechanism primarily enhances the fusion of text and image features, but the model can still rely on other mechanisms (e.g., MGCN) to some extent.

Finally, removing the LLM component caused a significant performance drop, with accuracy decreasing from 0.9526 to 0.9025, precision from 0.94 to 0.88, recall from 0.95 to 0.89, and F1 score from 0.94 to 0.88. This highlights the importance of the LLM in understanding user intent and generating personalized recommendations. The LLM's ability to process natural language and reflect on user feedback is critical for improving recommendation accuracy and user satisfaction.

In summary, the ablation study demonstrates that all components (MGCN, Cross-Attention, and LLM) contribute to the model's performance, with MGCN and LLM having the most significant impact. The full model, which integrates all components, achieves the best performance, underscoring the importance of their collaborative functioning.

*(b) Conclusion:* MGCN and LLM are the most critical components of the model, and their removal leads to a significant decrease in performance. The Cross-Attention Mechanism has a relatively minor impact on performance improvement, but it still positively contributes to the model's multimodal information fusion capability. The full model, which includes all components, achieves the best performance across all metrics, demonstrating the importance of the collaborative functioning of each component.

**Comparison with state-of-the-art methods**

We also compared our proposed model with two state-of-the-art baseline methods (LightGCN and DualGNN), with results as follows

*(a) Key Analysis:* Our Model versus LightGCN, our model outperforms LightGCN by 0.316 in accuracy and by 0.406 in F1 score (0.94 vs. 0.90). This indicates that our model significantly surpasses LightGCN in handling multimodal data and generating personalized recommendations.

Our Model versus DualGNN, our model exceeds DualGNN by 0.241 in accuracy (0.9526 vs. 0.9285) and by 0.330 in F1 score (0.94 vs. 0.91). This suggests that our model is superior in capturing user intent and integrating multimodal information compared to DualGNN.

Baseline Performance, the performances of LightGCN and DualGNN are similar, though DualGNN slightly outperforms LightGCN across all metrics. Nevertheless, our model significantly outperforms both baseline methods in accuracy, precision, recall, and F1 score.

*(b) Conclusion:* Our model significantly outperforms the latest baseline methods (LightGCN and DualGNN) in terms of accuracy, precision, recall, and F1 score. This indicates that our model has stronger performance in multimodal recommendation tasks, effectively handling complex user intents and multimodal data.

Summary analysis

Ablation Study,MGCN and LLM are the most critical components of the model, and their removal leads to a significant decline in performance. The Cross-Attention Mechanism has a relatively minor impact on performance improvement but still positively contributes to multimodal information fusion. The full model (with all components) performs best across all metrics, underscoring the importance of the collaborative functionality of each component.

Comparison with Baseline Methods,Our model significantly outperforms the latest baseline methods (LightGCN and DualGNN) in accuracy, precision, recall, and F1 score. This demonstrates that our model exhibits superior performance in multimodal recommendation tasks, effectively addressing complex user intents and multimodal data.

Experimental setup

This study aims to construct a multimodal dialogue system that determines user intent based on keywords provided by the user and offers corresponding services according to different intents. The specific tasks include chat dialogues and travel consultations, with the latter involving expert advice from various fields such as natural sciences, social sciences, humanities, engineering and technology, medical and health sciences, business, and management.

LightGCN is a streamlined yet powerful Graph Convolution Network (GCN) specifically designed for collaborative filtering recommendation tasks<sup>49</sup>. Due to its excellent performance, I integrated it with BILP and CILP to endow it with multimodal recommendation capabilities, enabling comparisons with my algorithm.

Hyperparameters

In our deep learning model, we carefully selected a set of hyperparameters to achieve optimal performance in text and image processing tasks. The choice of these hyperparameters was meticulously considered to maximize the model's performance for the given tasks. For the specific parameters of the models mentioned in this article, please refer to Table 3.

Firstly, we defined the maximum length of text sequences (Max\_sequence\_length) as 1000, reflecting the need to process text data containing rich information. To represent the text, we utilized a 300-dimensional embedding layer (Embedding\_dim) to ensure the retention of sufficient semantic information (Table 6).

In the design of the Convolutional Neural Network (Conv1D), we chose 100 filters (num\_filters) with a filter size (filter\_size) of 3. This configuration was carefully selected to maintain moderate complexity while capturing local features in the text. Additionally, we introduced a 50% Dropout (drop) to prevent overfitting, thereby enhancing the model's generalization capability.

The vocabulary size (vocab\_size) was set to 10000, balancing the richness of the vocabulary and the computational efficiency of the model. This choice was based on our analysis of the dataset to ensure an adequate vocabulary size to capture the diversity of the text.

For the optimizer selection, we adopted the Adam optimizer with an initial learning rate (lr) of 0.001. This choice was based on the widespread application of the Adam optimizer in training deep learning models and has been empirically proven effective for our task.

Finally, during the training process, Mean Squared Error (MSE) was employed as the loss function since our task is a regression problem. Accuracy was used as the metric to evaluate the model's performance.

Comparison of model effects

In the part of evaluation metrics,to assess the performance of our model, we employ the following metrics. Accuracy is used to measure the model's accuracy in the task of purpose identification, i.e., the proportion of correctly classifying user needs and purposes. In the part of model descriptions, Blip2 is a multimodal model excelling in natural language processing and image processing. We select Blip2 as a benchmark for comparison with existing state-of-the-art models<sup>51</sup>. In our dataset, the algorithm achieves an accuracy of 0.6541.

CLIP is a contrastive learning-based multimodal model with powerful text and image understanding capabilities. We use CLIP as a benchmark to evaluate our model's performance on text-image contrast tasks<sup>52</sup>. In our dataset, the model achieves an accuracy of 0.5561.

The initial version performs exceptionally well on our dataset, with an accuracy of 0.9519.

The cross-attention version achieves an accuracy of 0.9524 on our dataset.

The MGCN version attains an accuracy of 0.9526 on our dataset.

Comprehensive Comparison In the comprehensive comparison, the introduction of the cross-attention mechanism improves accuracy by 0.0005, and the introduction of MGCN improves accuracy by 0.0002. We also

| Parameter           | Value |
|---------------------|-------|
| Max sequence length | 1000  |
| Embedding dimension | 300   |
| Number of filters   | 100   |
| Filter size         | 3     |
| Dropout rate        | 0.5   |
| Vocabulary size     | 10000 |

Table 6. Model configuration.



| Model           | Accuracy | Precision | Recall | F1 Score | RMSE |
|-----------------|----------|-----------|--------|----------|------|
| Blip2           | 0.6541   | 0.64      | 0.66   | 0.63     | 0.23 |
| CLIP            | 0.5561   | 0.55      | 0.58   | 0.56     | 0.31 |
| Regular Version | 0.9519   | 0.94      | 0.96   | 0.95     | 0.08 |
| Cross Attention | 0.9524   | 0.93      | 0.94   | 0.93     | 0.07 |
| MGCN            | 0.9526   | 0.94      | 0.95   | 0.94     | 0.06 |

**Table 7.** Key performance indicators for different models.

train and evaluate comparative models such as Blip2, CLIP to compare with state-of-the-art multimodal models. Through a detailed analysis of the model’s performance on different tasks and validation of the experimental results“robustness and feasibility through cross-validation and actual user feedback, we can comprehensively evaluate the performance of our multimodal consultation model in the tourism consultation task. This allows us to determine its strengths and weaknesses relative to other models. These results will guide further improvement and optimization of the model to provide a better travel experience. Table 7 shows the key performance indicators of different models.

This table covers key performance indicators for five different models, including accuracy, precision, recall, F1 score, and root mean square error (RMSE). A preliminary data analysis is conducted for each model. Blip2 exhibits a high accuracy of 0.6541, but in comparison to other metrics, its precision and recall are relatively balanced. With an F1 score of 0.63, it performs well in balancing precision and recall. The RMSE of 0.23 indicates relatively small errors for regression tasks. CLIP has a relatively lower accuracy of 0.5561. Its precision, recall, and F1 score are 0.55, 0.58, and 0.56, respectively, suggesting limited performance in certain tasks. The RMSE of 0.31 may indicate significant errors in regression tasks. The Initial Version outperforms all models with a high accuracy of 0.9519. It boasts excellent precision (0.94) and recall (0.96), resulting in an impressive F1 score of 0.95. The RMSE of 0.08 signifies relatively small errors. Cross Attention achieves an accuracy of 0.9524, comparable to the Regular Version. It demonstrates good precision (0.93) and recall (0.94), yielding an F1 score of 0.93. The RMSE of 0.07 indicates minor errors. MGCN achieves an accuracy of 0.9526, slightly higher than Cross Attention. It has good precision (0.94), recall (0.95), and F1 score (0.94). With an RMSE of 0.06, it exhibits the smallest errors among all models. Overall, Initial Version, Cross Attention, and MGCN perform exceptionally well across all metrics, especially in terms of accuracy and F1 score. Blip2 shows a more balanced performance, while CLIP falls slightly short in accuracy and related metrics. Model selection should be based on specific task requirements and performance criteria.

In my research, I focus on specialized datasets such as the Guangdong Tourism Dataset and QK-Video, which differ significantly from classic recommendation system datasets like Movielens and Netflix. These specialized datasets possess characteristics such as content specificity, user behavior sparsity, and context dependency, posing challenges for traditional recommendation models in capturing sparse user feedback. The motivation behind my work is to address these recommendation challenges in specialized datasets, including sparsity, context dependency, and the need for recommendations tailored to specific content. Meanwhile, I highlight the limitations of existing recommendation models in capturing sparse user feedback, making them slow to adapt and learn. To address this, I propose a framework that combines advanced recommendation models with Low-Rank Models (LLM), aiming to quickly capture sparse user feedback through efficient representation learning and provide more accurate recommendations in a shorter time. Experimental results demonstrate that my recommendation+LLM framework performs excellently on both objective and subjective evaluation metrics, achieving impressive performance not only on public datasets but also exhibiting significant advantages on private datasets, showcasing its strong generalization capability and practicality.

Additionally, this paper leveraged the QK-video dataset<sup>53</sup>, a large-scale, multi-scene real-world dataset collaboratively released by Tencent and Westlake University, specifically tailored for evaluating recommendation systems<sup>53</sup>. Originating from Tencent’s two major information flow recommendation platforms, OK platform and QB platform, the dataset encompasses both articles and videos with varying degrees of overlap, boasting an expansive scale with approximately 5 million users and 140 million interaction records, including clicks, likes, and shares. The dataset further provides intrinsic user characteristics like age and gender, along with item features such as video types. It’s worth noting that the click behavior in the QK-video dataset follows a long-tail distribution, while session lengths are predominantly distributed between 0 and 20. Furthermore, we discussed the Tenrec dataset, an extension of the QK-video dataset, tailored for evaluating a diverse array of recommendation tasks, including click-through rate prediction, session-based recommendation, multi-task recommendation, and more. By publicly releasing the dataset, code, and leaderboards, Tenrec aims to catalyze further advancements in the recommendation system research community.

This study comprehensively demonstrate its core contributions through experiments. Firstly, the MGCN model excels on both public and private datasets, particularly on the Guangdong Tourism Dataset, QK-Video, and QB-Video datasets, where its Recall metrics significantly outperform other comparative models. Secondly, the introduction of a Large Language Model (LLM) enhances the model’s performance in cold start and sparse user feedback scenarios. The application of LLM across multiple roles, including Actor, Evaluator, Self-reflection, and Memory, bolsters the decision-making quality and personalized recommendation capabilities of the model. Through in-depth analysis of user-system interactions, LLM extracts user preferences and subtle interaction nuances. Lastly, we have constructed a novel dataset encompassing information on major tourist cities in Guangdong Province. This dataset furnishes essential inputs and labels for the research on multimodal attraction

| Dataset         | Method          | Recall@5 | Recall@10 | Recall@50 |
|-----------------|-----------------|----------|-----------|-----------|
| Tourism Dataset | Clip            | 0.3423   | 0.4867    | 0.6682    |
|                 | Blip2           | 0.4531   | 0.5763    | 0.7283    |
|                 | Cross attention | 0.5520   | 0.8522    | 0.9524    |
|                 | MGCN            | 0.5524   | 0.6524    | 0.9524    |
| QK-Video        | Clip            | 0.4863   | 0.5974    | 0.7104    |
|                 | Blip            | 0.5143   | 0.63967   | 0.76931   |
|                 | Cross attention | 0.5866   | 0.6077    | 0.8605    |
|                 | MGCN            | 0.6556   | 0.7677    | 0.9559    |
| QB-Video        | Clip            | 0.46631  | 0.5882    | 0.7563    |
|                 | Blip            | 0.5021   | 0.63854   | 0.77862   |
|                 | Cross attention | 0.5766   | 0.6874    | 0.8705    |
|                 | MGCN            | 0.6856   | 0.7977    | 0.9059    |
| QK-article      | Cross attention | 0.5864   | 0.6976    | 0.8984    |
|                 | MGCN            | 0.6857   | 0.7056    | 0.9161    |

**Table 8.** Experimental results on Guangdong tourism dataset.

rating frameworks, thereby enhancing the operational feasibility and accuracy of the model. This unique dataset not only broadens the model's applicability but also provides a valuable resource for researchers to delve deeper into multimodal data processing and recommendation system applications. Table 8 shows the experimental results of different data sets on the Guangdong tourism data set.

**Module function and performance analysis**

The model comprises a MGCN, Residual Network (ResNet), and Cross Attention Mechanism. Cross Attention Mechanism, in a multi-agent system, collaborative cooperation and information sharing are crucial. The introduction of a collaborative attention mechanism enables effective information transfer and collaboration among different components. This mechanism ensures that each component can focus on important tasks while sharing critical information to optimize the overall system performance. The incorporation of Cross Attention into the model primarily addresses the processing requirements for multimodal data. Intelligent decision support systems may need to handle various data types such as text and images. Through Cross Attention, key features can be extracted from images, aiding the system in better understanding and processing image information, such as scenic spot pictures. Multi-View Convolutional Neural Network, the Multi-View CNN extends the capabilities of traditional CNN, particularly suitable for handling multimodal data. In this context, both scenic descriptions and pictures can be considered as different “views”, and the Multi-View CNN effectively merges this information, providing a more comprehensive data representation. This contributes to enhancing the overall performance of the system and the quality of intelligent consultations. ResNet is widely employed in deep learning, with its core idea of addressing the vanishing gradient problem by adding short connections (or skip connections), allowing the construction of very deep neural networks. In intelligent decision support models, especially in multi-agent systems, deep networks contribute to learning complex decision strategies and recommendation generation. The introduction of ResNet helps overcome gradient issues when training deep models, thereby improving system performance.

Integrating technologies such as MGCN, Cross Attention, and ResNet into the tourism advisory system brings various advantages and addresses several challenges in the field. MGCN efficiently handles multimodal data, including text and images, facilitating a better integration and understanding of diverse modalities. This is particularly crucial for tourism advisory systems, as users may provide textual descriptions, images, or a combination of both. The Cross Attention mechanism enhances the model's understanding of inter-modality relationships, empowering the model to concurrently consider both textual and visual information, thereby comprehensively meeting user needs. ResNet serves as powerful models for processing image data. In tourism advisory scenarios where users may submit images of attractions, employing these models allows the system to extract more advanced and meaningful features from the images, providing richer information for recommendations and consultations. The residual network structure of ResNet aids in mitigating the vanishing gradient problem during the training of deep neural networks, making the network more amenable to training and optimization. Multi-View Neural Networks contribute to handling diverse modal data, preserving their respective information and integrating them. This is particularly crucial for synthesizing textual and visual information provided by users. The Self-Reflection mechanism enables the model to consider its own state during decision-making, enhancing the system's self-awareness and adjustment capabilities. This contributes to improving the accuracy of recommendations and user satisfaction. Pretraining text with BERT facilitates a better understanding and processing of descriptions of attractions within the system. Pretraining images with ResNet aids in extracting advanced features from images, offering the system additional visual information. By integrating these advanced technologies, the tourism advisory system can comprehensively and intelligently handle user queries, providing more accurate and personalized recommendations. This integration aims to enhance user experience and overall system performance.

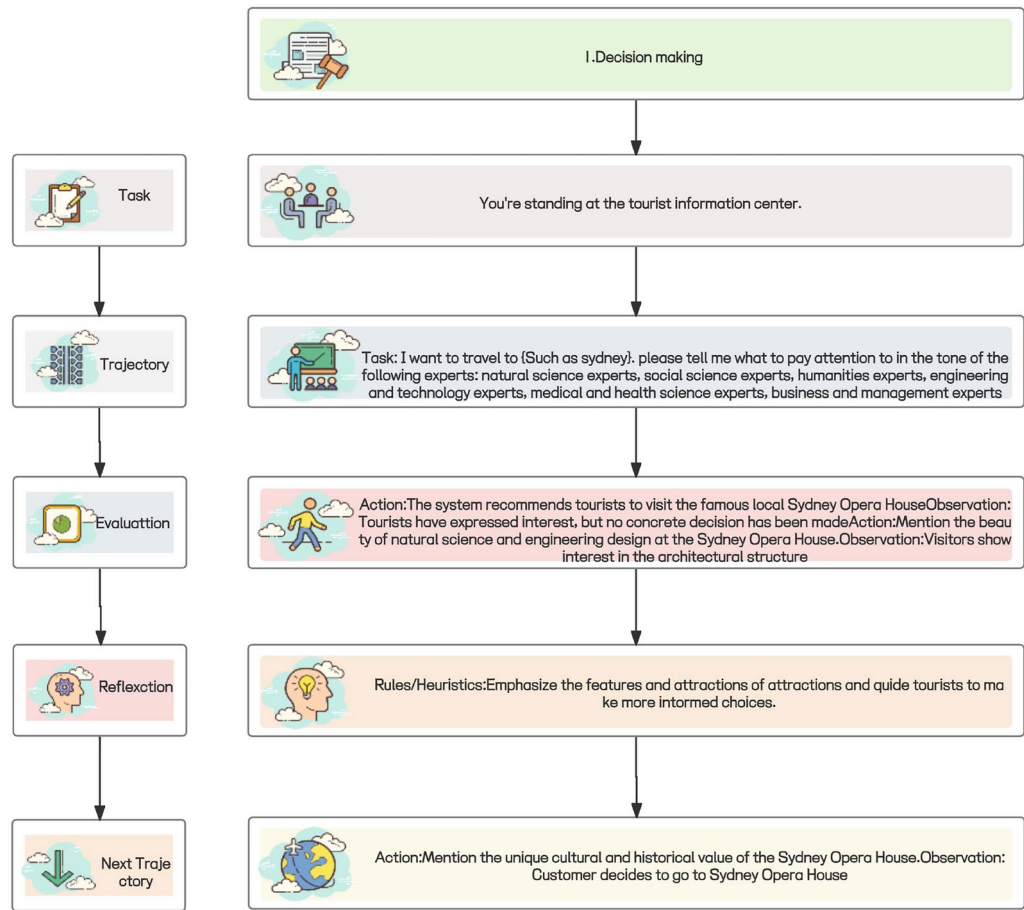


Fig. 4. The flowchart of reflection.

| Group        | Pass1 accuracy | Pass2 accuracy |
|--------------|----------------|----------------|
| Experimental | 68%            | 91%            |
| Control      | 65%            | 72%            |

Table 9. Results of the reflexion mechanism experiment.

In this code segment, my focus is on text and image processing to address challenges within the tourism advisory system. For textual data, Tokenizer is employed for word segmentation, followed by the utilization of the Embedding layer to transform the text into vector representations. Additionally, the MGCN is applied to extract crucial features from the textual data. To enhance model performance, pre-trained word vectors are incorporated using GloVe embeddings. For image data, I introduced ResNet (Residual Neural Network) to handle scenic images, utilizing the MGCN for convolutional operations and ultimately applying GlobalMaxPooling1D to achieve global pooling on the convolutional results. This aids in extracting advanced features from images, providing more information for recommendations and consultations.

Reflexion improves interactive performance

The function of this module is to enhance LLM’s recognition of the user’s purpose and improve the user’s experience during the conversation with the user. Through this module, LLM will conduct a simple reinforcement learning of its own conversation with the user and constantly prompt LLM according to the user’s language, so as to more accurately recognize the user’s intention and improve the user experience.

Reflexion ablation experiment

This paper designed the following experiment, which consists of two groups: the experimental group and the control group. The experimental group utilizes the reflexion mechanism, while the control group does not. The purpose of the experiment is to evaluate the impact of the reflexion mechanism on the model’s performance. Fig. 4 is the flowchart of reflection. The experiment process is Participants in the experimental group carry out a travel recommendation task, and the model uses the reflexion mechanism during the task execution. Firstly,

the model queries experts from different fields based on the participant's stated travel purpose and preferences to obtain travel recommendations. Then, based on the expert advice and participant responses, the model engages in decision-making reasoning using the reflexion mechanism and selects the next course of action. Finally, the model provides the ultimate travel recommendation. As shown in Table 9, the pass1 accuracy for the experimental group is 0.68, and the pass2 accuracy is 0.91. Participants in the control group perform the same travel recommendation task as the experimental group, but the model does not utilize the reflexion mechanism during the task execution. The model directly engages in decision-making reasoning based on expert advice and participant responses, selecting the next course of action. The pass1 accuracy for the control group is 0.65, and the pass2 accuracy is 0.72. Table 6 is the results of the reflection mechanism experiment.

#### *Decision making*

For decision-making, in the task part, users are planning a trip with the objective of choosing a destination recommended by experts. In the trajectory part, six expert agents responded to the user's inquiries simultaneously based on their expertise. The natural sciences expert suggested Iceland for its natural landscapes. The social sciences expert recommended Japan for its cultural experiences. The humanities expert proposed Italy for its historical sites. The medical and health sciences expert advised Switzerland for its relaxation and health facilities. The engineering and technology expert favored Singapore for its advanced technology and architecture. The business and management expert suggested New York for its business and financial center.

In the evaluation part, users responded positively or negatively. Despite the natural sciences expert not being online at the time, the user continued to consider advice from other experts. In the reflection part, users contemplated each destination's unique attractions and their personal interests based on the advice from different experts. For the next trajectory, users ultimately made a decision that balanced personal interests with the experts' recommended travel destination. Specifically, the process is as shown in Fig. 5.

#### *Reasoning task*

For the reasoning task,

In the task part, a tourist is planning their vacation next month and seeks evaluations and recommendations about travel destinations, aiming to choose a place that aligns with personal interests and receives expert endorsement.

In the trajectory part, the tourist asked various experts for evaluations of a destination: the tourist rates it 8/10 for its scenic beauty with a bridge spanning the sea offering views and opportunities to study marine life; the natural sciences expert rates it 7/10, seeing the bridge linking regions as fostering economic and cultural exchange; the social sciences expert rates it 7/10, viewing the bridge as a symbol of human ingenuity; the humanities expert rates it 6/10, appreciating the fresh air on the bridge; the engineering and technology expert rates it 9/10, praising the modern engineering of the bridge; the business and management expert rates it 7/10, believing the bridge's construction promotes economic growth; the medical and health sciences expert rates it 8/10, praising the bridge's health and relaxation environment.

In the evaluation part, the tourist feels the experts' evaluations are not quite suitable and requests reconsideration.

In the reflection part, the system thanks the tourist for their feedback and asks for specifics on dissatisfaction with the evaluations or expert opinions to improve recommendations. The tourist points out that the natural sciences expert's rating is too high and the engineering and technology expert's rating is also slightly high.

The experts provide adjustment reasons: although the bridge offers beautiful scenery, biodiversity and research opportunities are limited, so the natural sciences expert's rating is adjusted to 7/10. Despite excellent engineering, maintenance challenges and environmental impact reduce the engineering and technology expert's rating to 8/10.

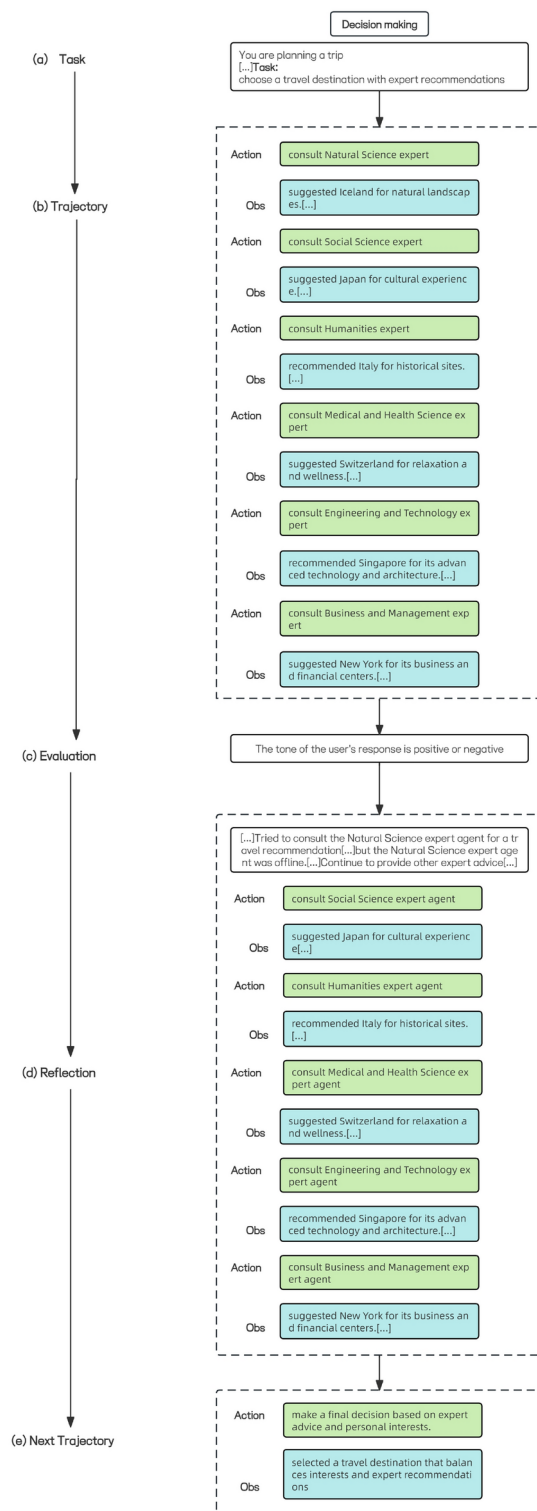
For the next trajectory, the tourist accepts the adjusted ratings and expresses satisfaction. The system suggests further optimizing the tourist's travel experience and asks if they would like more information about this attraction. Specifically, the process is as shown in Fig. 6.

#### *Travel plan*

In the travel plan task, the tourist is preparing to arrange a trip but has not yet determined specific plans. They seek assistance from experts to plan a perfect travel itinerary.

In the task section, the tourist seeks expert help to craft a flawless travel plan. In the trajectory section, the tourist inquires about advice and ratings for Shantou Old Town, expressing a love for ancient architecture. Throughout the process, experts provide ratings and recommendations.

Then in the evaluation section, based on the tourist's interests and schedule, they choose the most suitable travel destination and develop a corresponding travel plan. In the reflection section, the natural sciences expert rates Shantou Old Town highly for its exceptionally beautiful natural environment, perfect for nature enthusiasts. The social sciences expert notes the town's rich history and culture, suggesting that exploring its streets provides deep insights into local traditions and history. The humanities expert praises the town's many unique ancient buildings, ideal for architecture enthusiasts to appreciate and study. The medical and health sciences expert recommends Shantou Old Town for walking but advises on sunscreen and hydration for a comfortable sightseeing experience. The engineering and technology expert acknowledges the town's aging infrastructure but finds its architectural techniques and historical value very intriguing. The business and management expert highlights the town's numerous unique shops and restaurants, offering local cuisine and a vibrant commercial atmosphere. In the next trajectory section, the tourist expresses gratitude for the experts' advice, looks forward to the trip, and indicates no further need for suggestions. Specifically, the process is as shown in Fig. 7.

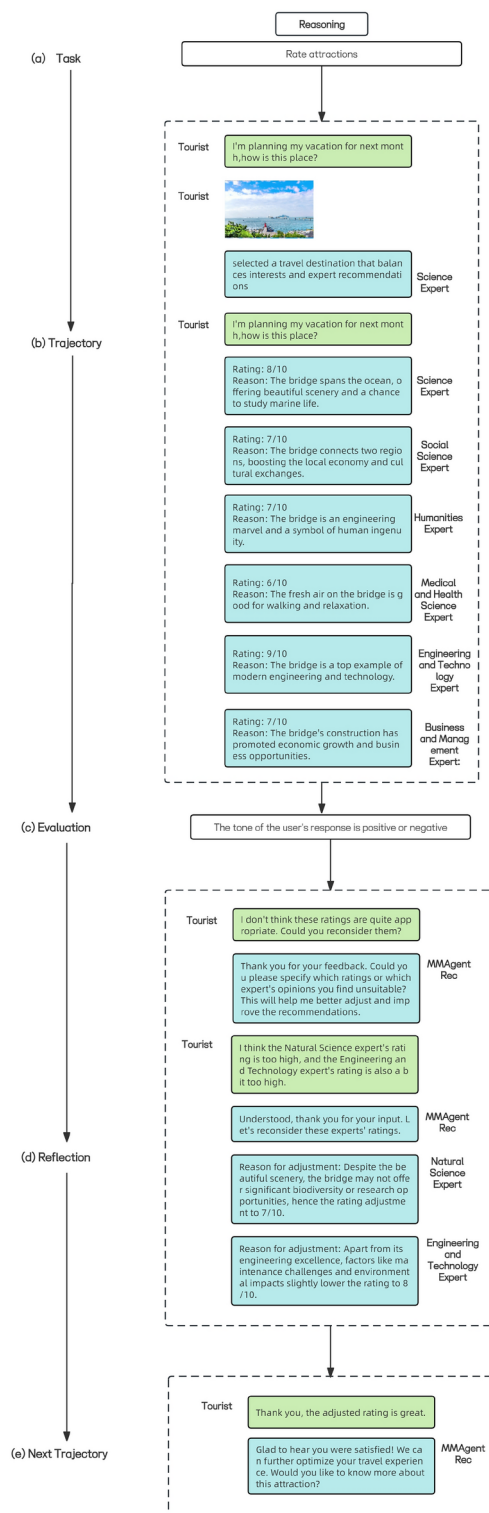


**Fig. 5.** Reflection works on decision-making.

## Human study

Ethical Statement as fellow.

All human subject experiments and/or the use of human tissue samples reported in this manuscript were conducted in full compliance with the relevant ethical guidelines and regulations. The experimental protocols were reviewed and approved by licensing committee of Fenglang Technology Co. All experiments were performed in accordance with the prescribed ethical standards. Furthermore, informed consent was obtained from all participants and/or their legal guardians prior to participation in the study. Detailed information



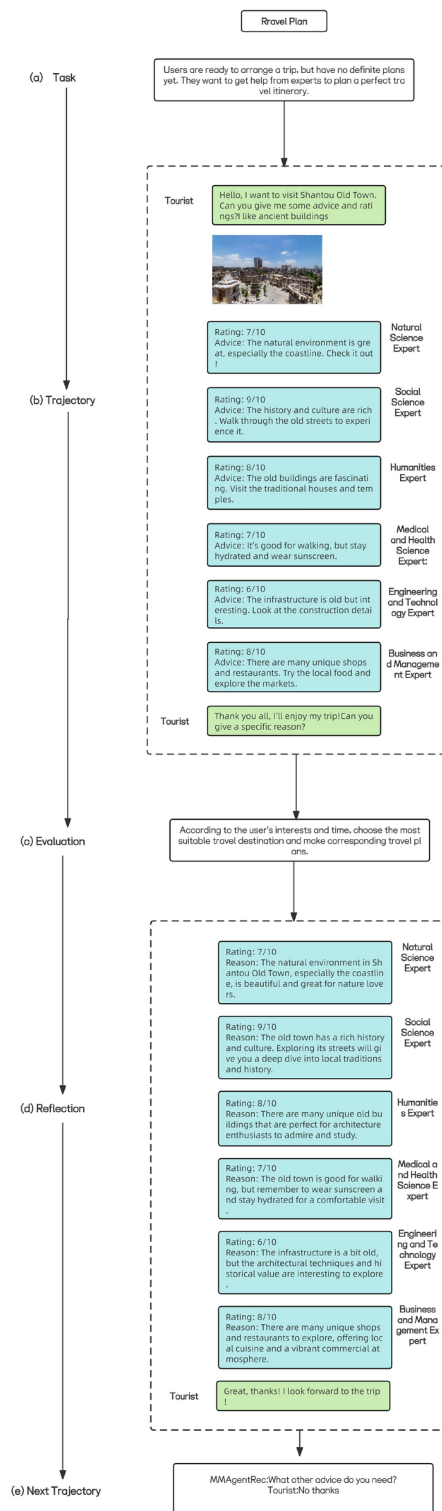
**Fig. 6.** Reflection works on reasoning.

regarding the approval process, including the approving institutions and relevant committee, is provided in the “Methods” section of the manuscript.

All methods were conducted following the relevant guidelines and regulations. The experiments were approved by the designated institution and/or licensing committee. Informed consent from all participants and/or their legal guardians was obtained. This should fulfill the requirements of the approval committee.

To validate the performance of the system, we conducted a human study. The research design includes the following steps:





**Fig. 7.** Reflection works on travel plan tasks.

**Participant recruitment:** We recruited participants from a variety of ages and backgrounds to ensure that the findings were representative.

**Experimental process** is participants were asked to conduct a series of conversations with the system, involving issues and tasks in different fields. Their inputs and the system's responses are recorded for subsequent analysis.

**Feedback collection** is that after the experiment, we collected feedback from participants, including their satisfaction with the system's performance, understanding, and suggestions for improvement. This feedback will be used to evaluate the effectiveness of the system and user experience.

*Experiments in adapting models to user preferences*

We designed this experiment through a questionnaire survey. A total of 1000 questionnaires were distributed, of which 500 were male Female: 450 Others: 50 people

*Questionnaire sample*

Dear Participant,

Thank you for participating in our research! Your valuable feedback is crucial for improving our conversational system. Please take some time to complete the following survey to help us better understand the system's performance and user experience.

*Questionnaire data and data analysis of questionnaire data***Analysis of user age**

In terms of demographic characteristics, we observed that the age distribution of participants was wide, with an average age of 34 years old, and the youngest and oldest participants were 16 and 78 years old, respectively. In terms of gender, male and female participants were basically balanced, while the proportion of participants of other genders was small. The occupational distribution showed that the system mainly attracted students and working people, while freelancers and participants of other occupations were relatively small, which showed that the system met the needs of people of all ages and occupational backgrounds and had the potential for practical implementation.

**Analysis of user experience**

Regarding user experience, most participants said that they had used similar systems before, accounting for 0.65

**Analysis of dialogue system experience**

Overall, the participants' satisfaction with the system was mainly concentrated in the satisfied and general levels, accounting for 0.60, while 0.25 indicated dissatisfied and very dissatisfied. In terms of system understanding and response accuracy, 0.30 participants indicated full understanding, 0.40 indicated some understanding, 0.20 indicated little understanding, and 0.10 indicated no understanding at all. These results indicate that the system exhibits good practicality and ease of use in terms of user experience, meets human usage needs, and has the potential for practical implementation.

**System function analysis**

The evaluation of the chat module of the system is more scattered, with 0.25 participants believing it is excellent, 0.35 believing it is good, and 0.10 believing it is relatively poor or very poor. For the tourism consultation module, 0.30 participants believed it fully met expectations, 0.35 believed it generally met expectations, and 0.10 believed it did not meet expectations well or did not meet expectations at all. These evaluations reflect the actual performance of the system under different functional modules, and also highlight the system's usability and user satisfaction, proving that the system meets human usage needs and has the potential for practical implementation.

**Willingness to participate in future research**

Regarding their willingness to participate in similar research in the future, 0.60 of the participants expressed their willingness to participate, indicating that they showed a strong interest in system improvement and future research. This further proves the practicality and user acceptance of the system, and provides strong support for future system improvement and further research.

**In-depth data analysis**

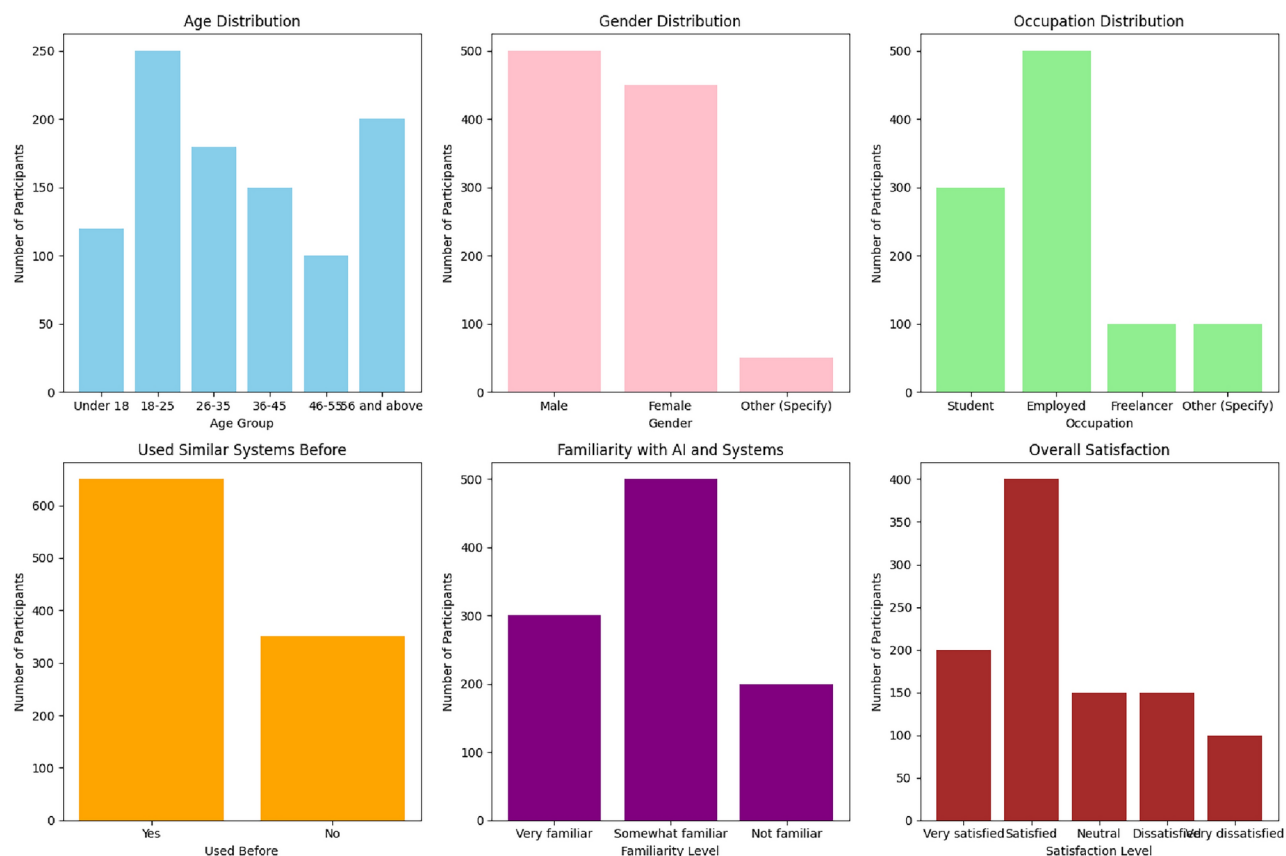
At a glance, this looks like what is shown in Fig. 8. According to data analysis, there may be a correlation between age and satisfaction. The mean age is about 1.7 and the variance is about 0.61, while the mean satisfaction is about 0.7143 and the variance is about 0.2041, indicating that the younger the age, the higher the satisfaction may be, but more in-depth analysis is needed. In addition, there may be a correlation between usage experience and system understanding. The mean usage experience is about 2.0 and the variance is about 0.5714, and the mean system understanding is about 0.5 and the variance is about 0.1429, indicating that the richer the usage experience, the higher the system understanding may be.

**Conclusion**

This paper focuses on constructing an innovative multimodal attraction rating framework, utilizing both textual and visual information to comprehensively assess the attractiveness and quality of tourist attractions. The framework introduces and MGCN to handle multimodal data, allowing the model to synthesize information from different data sources for a more comprehensive evaluation in travel planning and consultation.

Another significant contribution lies in the incorporation of interdisciplinary expert recommendations. Through prompt a Large LLM, representing experts from various fields including natural sciences, social sciences, humanities, engineering and technology, medical and health sciences, and business and management, the model provides users with interdisciplinary travel advice, enriching the diversity and adaptability of recommendations.

The prompt of the Large Language Model serves multiple roles, such as Actor, Evaluator, Self-reflection, and Memory, positioning the LLM as a core component of a multi-agent system. This enhances the user experience in travel consultation and dialogue, while simultaneously improving decision-making quality.



**Fig. 8.** Statistics on survey option selections.

The construction of a new dataset supports our research, containing key information about attractions in major tourist cities in Guangdong Province. This dataset, comprising 9004 entries per column, transforms attraction descriptions and images into vectors using pre-trained BERT and ResNet50 models, respectively, providing essential inputs and labels for model training and enhancing the operational feasibility of our research.

In terms of model results, a detailed evaluation and comparison of different versions of multimodal models, including Blip2, CLIP, Regular Version, Cross Attention, and MGCN, was conducted. These versions achieved significant accuracy on our dataset, with improvements of 0.0005 and 0.0002 introduced by the cross-attention mechanism and MGCN, respectively. Comparative analysis with state-of-the-art multimodal models like Blip2, CLIP was performed. Through a thorough examination of the model's performance across various tasks and validation of results through cross-validation and real user feedback, we comprehensively evaluated the performance of our multimodal consultation model in the tourism consultation task. These findings will guide further improvements and optimizations to provide an enhanced travel experience.

### Data availability

The data supporting the findings of this study are openly available in the Tenrec Benchmark Dataset repository at <https://openreview.net/forum?id=PfuW84q25y9>. This dataset includes a comprehensive collection of benchmarks for recommender systems, as described in the paper “Tenrec: A Large-scale Multipurpose Benchmark Dataset for Recommender Systems” by Guanghu Yuan et al. The data used for this study are publicly accessible in the following GitHub repository: Tourism-data-set. This dataset contains data for tourism research, manually crawled and organized by the authors of this study. The specific address is <https://github.com/971851309/Tourism-data-set>

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## Author contributions

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## Additional information

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